

CourtEdge — An NBA +EV Betting Analytics Platform

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GitHub repository: <https://github.com/Kaiweiyan/CourtEdge>

Live demo: <http://140.112.30.184:9999>

One-paragraph pitch. CourtEdge is an analytics platform for **serious recreational NBA bettors** in the US legal-sports-betting market. It does **not** sell “guaranteed picks.” It ingests NBA stats and multi-sportsbook odds, runs a *calibrated* win-probability model, and helps the user bet smarter through four defensible features: **edge detection** (model probability vs. market-implied probability), **line shopping** (best price across books), **CLV tracking** (Closing Line Value — the real long-term skill metric), and **Kelly bankroll sizing**. The monetized asset is the *refined data product*: odds + model probabilities + edge, surfaced in one screen.

Why this framing is defensible. The NBA market is highly efficient, so a claim of “we beat the market” is hard to defend. CourtEdge instead promises to help users *get the best price, track their CLV, and size bets correctly* — which provably saves/makes money and sidesteps the efficiency trap — while the calibrated model stays the technical centerpiece. We make **no profit guarantee**; we report calibration and CLV honestly.

1 Target Customer

Primary segment. US-based *serious recreational* NBA bettors: people who already bet regularly (accounts at two or more sportsbooks), bet for profit rather than pure entertainment, but are not professional syndicates. After the US Supreme Court struck down PASPA in May 2018, legal sports betting expanded to roughly 38–39 states plus DC, creating a large, growing, English-speaking market essentially from scratch.

The job they are trying to do today.

- Manually check odds across DraftKings / FanDuel / BetMGM / etc. to find the best line.
- Guess whether a line is “good value” without a probabilistic baseline.
- Track bets in a spreadsheet — most do not track CLV at all.
- Size bets by gut feel rather than bankroll math.

Current workarounds. Spreadsheets, free Reddit picks, scattered odds-comparison sites, and expensive incumbents (OddsJam, Action Network, Unabated) at roughly \$10–150/mo.

Why CourtEdge is better than the status quo. One place that unifies model edge + line shopping + CLV tracking + Kelly sizing, with a *transparent, auditable* methodology (no “trust me” black box). It is cheaper and more focused than the premium incumbents and positions on honesty rather than hype.

Secondary segment (future). Sell the model-probability + edge **data feed via API** to other tool builders (B2B2X). Out of scope for v1 but noted for go-to-market.

2 Evidence of Demand & Willingness to Pay

The rubric rewards the *process* from a vague idea to a defensible demand claim, not just the conclusion. The full write-up and raw artifacts live in `research/demand_evidence.md`; this section summarizes the journey.

Step 1 — Hypothesis. Serious recreational bettors will pay a monthly subscription for a tool that (a) finds the best price across books, (b) flags potentially mispriced lines, and (c) tracks CLV and sizes bets correctly — *without* the scammy “guaranteed picks” framing.

Step 2 — Is the market real? (public-data analysis)

- **Legalisation tailwind:** PASPA struck down May 2018; legal betting now in ~38–39 states + DC — a newly created, large addressable market.
- **Market size:** US commercial sports-betting revenue is on the order of \$10B+/year (American Gaming Association); even a small analytics attach-rate is a meaningful TAM.
- **Engaged communities (proxy for the segment):** r/sportsbook (~600k+ members) with daily line-shopping and bad-beat threads; r/dfsports, Discord groups, and Twitter/X “betting model” accounts.
- **Search intent & review mining:** Google Trends for +EV betting, line shopping, NBA betting model, closing line value; and app-store reviews of incumbents mined for recurring praise (“found me +EV”) and complaints (“too expensive,” “cluttered”) → unmet needs.

Step 3 — Will they pay? (competitor pricing as a WTP anchor). Existing paid tools prove willingness to pay (approximate public tiers):

Product	What it does	Approx. price
OddsJam	+EV / arbitrage / line shopping	~\$75–150 / mo
Unabated	sharp lines, no-vig, line shopping	~\$50+ / mo
Action Network (PRO)	odds, tracking, picks	~\$8–13 / mo
Outlier.bet	player props / +EV	freemium + sub

The market spans a cheap tracking tier (~\$10) to a premium +EV tier (~\$75–150). CourtEdge’s wedge is a focused, transparent, **mid-price (~\$15–25/mo)** +EV + CLV tool that undercuts the premium incumbents while doing the few things bettors value most.

Step 4 — Primary research (survey instrument). A short questionnaire (`research/`) targets 5–15 bettors: number of sportsbook accounts and whether they line-shop every bet; whether they track CLV (expected to be low — the gap we fill); how they size bets; what they have paid for tooling; and a van-Westendorp-style price test at \$10/\$20/\$30/mo. Responses are summarized as % who line-shop, % who track CLV, and the price point with the best yes-rate.

Step 5 — Non-monetary value (time saved). Manually checking ~4 books × ~6 games/night is several minutes/night of comparison; CourtEdge collapses that to one screen. Minutes saved × nights/season complements the monetary WTP estimate.

Conclusion. A large, newly legal, engaged market already pays \$10–150/mo for adjacent tools, yet most bettors do not track CLV and line-shop inconsistently. That gap, plus a transparent, no-hype, mid-price position, is a defensible wedge. The honest risk (section 3) is a weak data moat and the efficiency of the market itself.

Honesty note on figures. Market-size, state-count, subscriber and pricing numbers above are publicly reported approximations used to size the opportunity. Each is flagged [verify] in the repository’s demand-evidence write-up ([research/](#)) and should be re-checked against a dated primary source before external use.

3 Go-to-Market Difficulties

- **Market efficiency / trust.** NBA closing lines are near-optimal, so we must *not* promise profit. Trust is earned via a transparent, auditable CLV track record and open methodology — the differentiator from the “tout” scam industry.
- **Legal & compliance.** CourtEdge is **analytics only** — **it never accepts wagers**, which keeps it out of gambling-operator licensing. It must still respect state-by-state advertising rules, affiliate disclosure, responsible-gambling messaging, age-gating (21+), and each book’s Terms of Service.
- **Data-acquisition cost & risk.** `nba_api` is unofficial and rate-limited; The Odds API free tier is small (~ 500 req/mo), so polling for line movement gets expensive at scale — cost scales with $\text{polling frequency} \times \text{games} \times \text{books}$.
- **Cold-start.** Historical seasons are needed to backfill before the model is credible, and a CLV track record needs time to accumulate.
- **Competition & moats.** Incumbents are well-funded and the underlying data is largely public — the moat is weak. We are honest about this: the edge is focus, transparency, and price, not data exclusivity.
- **Unit economics.** At 100 users the binding cost is the odds-API tier + a single small VM (the demo runs on one workstation); at 1,000 users odds polling and DB load dominate, motivating caching, a tapering poll cadence, and a paid odds tier. Break-even is driven by odds-API spend, which is controllable by polling discipline.

4 System Design

4.1 Architecture overview

The system follows the course’s canonical flow: **ingestion** $\rightarrow$ **storage** $\rightarrow$ **processing** $\rightarrow$ **delivery**. Figure 1 shows the end-to-end pipeline with the file that implements each stage.

4.2 Data sources

Source	What	Access / risk
<code>nba_api</code>	game logs, play-by-play	free wrapper; unofficial, rate-limited
The Odds API	moneyline across US books	REST; free tier ~ 500 req/mo

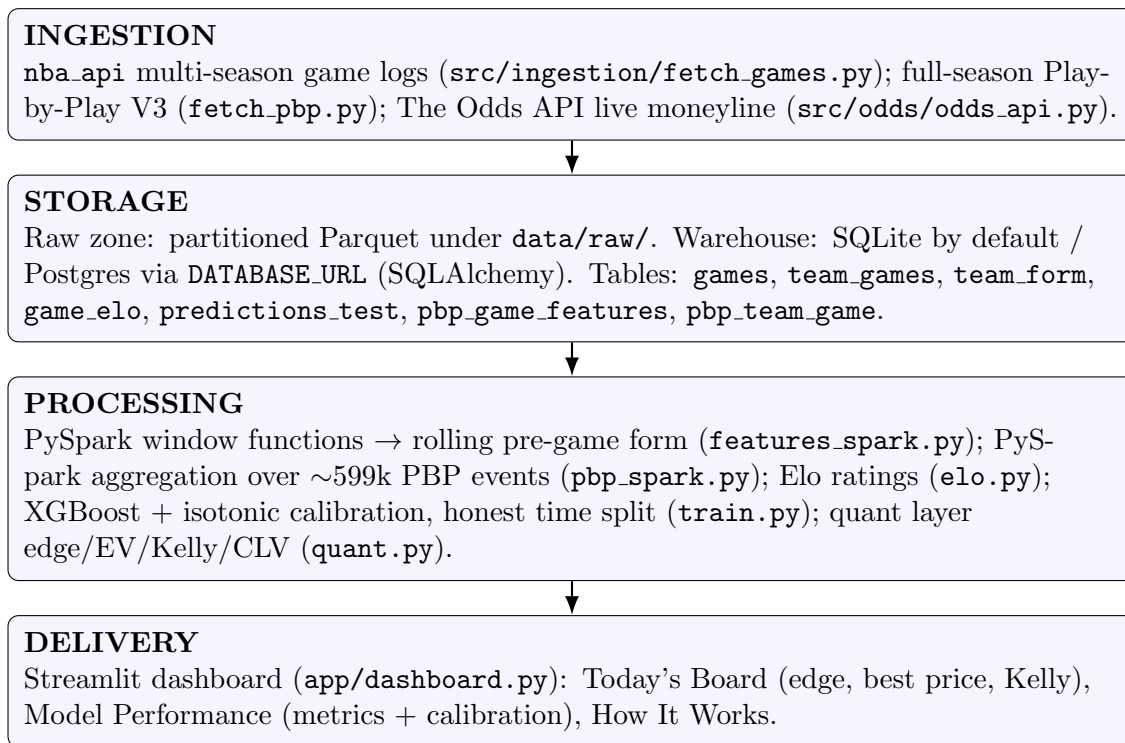


Figure 1: CourtEdge end-to-end architecture.

We ingest five recent seasons (2020–21 . . . 2024–25): **5,995 games** and **12,000 team-game rows**. For the big-data processing demo we ingest the **full 2023–24 play-by-play: 598,705 event rows across 1,230 games** (~486 events/game).

4.3 Storage — why these tools fit the scale

The raw zone is columnar **Parquet**, partitioned by season — efficient for Spark reads and the “data lake” story; the documented production path swaps it for MinIO/S3. The **warehouse** holds curated, query-served tables; we use SQLite by default for zero-friction reproducibility, and SQLAlchemy makes the switch to PostgreSQL a one-line `DATABASE_URL` change. Play-by-play stays in Parquet (too row-heavy for a transactional store) and is processed by Spark.

4.4 Processing — the legitimate big-data core

PySpark (running `local[*]`) builds rolling pre-game team-form features with *window functions* partitioned by team and ordered by date, using `rowsBetween(-N, -1)` so every feature uses only the *previous* N games (no leakage). The same DataFrame code aggregates the **598,705-row** play-by-play table into per-game and per-team descriptors (lead changes, max margin, OT flag, FG%, threes) — the genuinely distributed workload; identical code scales to a cluster and to all-time PBP (~10M+ rows). **Elo** ratings are computed sequentially in pandas (each game updates ratings, with season-to-season regression toward the mean).

Model. An **XGBoost** classifier over ten features (Elo expectation and difference, rolling net-rating, points, and win-rate differentials, rest days, back-to-back flags) with a strictly time-ordered

split: train on the oldest games, calibrate on a validation slice with **isotonic regression**, and test on the most recent season. Calibration is essential because edge = model prob – market-implied prob is only meaningful when the model probability is well calibrated.

4.5 The quant layer (exact formulas — this is the product)

American odds o convert to **implied probability** and **decimal odds** as

$$\text{imp}(o) = \begin{cases} \frac{100}{o + 100}, & o > 0 \\ \frac{-o}{-o + 100}, & o < 0 \end{cases} \quad d(o) = \begin{cases} \frac{o}{100} + 1, & o > 0 \\ \frac{100}{-o} + 1, & o < 0. \end{cases}$$

A two-way market is **de-vigged** to fair probabilities by $p_A^{\text{fair}} = \text{imp}_A / (\text{imp}_A + \text{imp}_B)$. With model probability p and $b = d(o) - 1$:

$$\underbrace{\text{edge} = p - \text{imp}(o)}_{\text{displayed}}, \quad \underbrace{\text{EV} = pb - (1 - p)}_{\text{per unit staked}}, \quad \underbrace{f^* = \frac{bp - (1 - p)}{b}}_{\text{Kelly fraction}}.$$

We bet only when $\text{EV} > 0$, stake $0.25 \times$ Kelly capped at 5% of bankroll, and measure **CLV** = $\text{imp}(\text{closing}) - \text{imp}(\text{taken})$ (positive $\Rightarrow$ you beat the close). All functions are pure and unit-tested in `src/model/quant.py`.

4.6 Delivery

A **Streamlit** dashboard serves three pages: *Today's Board* (model probability vs. market, best price across books, edge, EV, Kelly stake), *Model Performance* (out-of-sample metrics + calibration curve — the trust layer), and *How It Works*. When an Odds API key is set the board scores live upcoming games; off-season or keyless it falls back to a clearly labelled *sample board* built from real held-out predictions, so the UI always renders and the edge math stays real. The app is deployed on a school workstation (see page 1).

4.7 Results (held-out most-recent season, out-of-sample)

The test set is the 897 most recent games, never seen in training or calibration.

Model	n	Accuracy	Log loss	Brier	AUC
Elo baseline	897	0.654	0.616	0.214	0.723
XGBoost (raw)	897	0.678	0.609	0.210	0.731
XGBoost (calibrated)	897	0.669	0.608	0.210	0.728

Honest read. The NBA market is highly efficient, so a calibrated model lands around 65–68% accuracy / $\text{AUC} \approx 0.73$ and only *modestly* beats Elo. The top features are the Elo win-expectation (≈ 0.31 importance), Elo difference (≈ 0.21), and rolling net-rating differential (≈ 0.07) — i.e. strength + recent form, as expected. We make **no** profit guarantee; the product value is surfacing the *occasional* mispriced line, line-shopping the best price, and correct (Kelly) sizing. The north-star skill metric is CLV. The calibration curve (fig. 2) shows the calibrated model tracks the diagonal closely, which is what makes the edge comparison trustworthy.

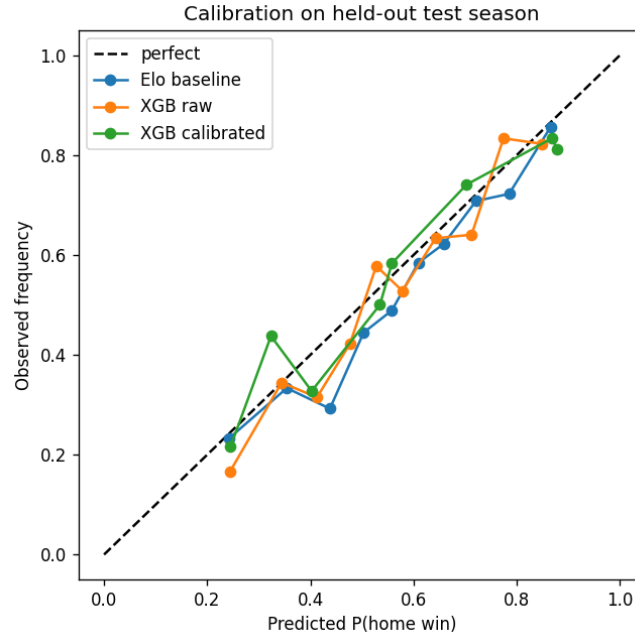


Figure 2: Reliability curve on the held-out test season (closer to the diagonal is better).

4.8 Honest scoping and the scaling path ($10\times$ / $100\times$)

What was built is a complete *thin slice* end-to-end. Deliberately descoped items map directly to the production architecture: SQLite $\rightarrow$ Postgres + MinIO/S3; Spark `local[*]` $\rightarrow$ a cluster (YARN/K8s) over all-time play-by-play ($\sim 10\text{M}+$ rows); Streamlit $\rightarrow$ FastAPI + Next.js with auth; and a daily Airflow DAG plus a Kafka odds stream for live line-movement and automated CLV capture. Because the Spark code is plain DataFrame/window-function code, it scales unchanged.

Reproducibility. The repository commits the built warehouse (`data/courtedge.db`) and model artifacts (`data/artifacts/`) so a fresh clone serves in ~ 1 minute via `requirements-serve.txt` (no Spark/Java). A full rebuild is one command — `bash run_pipeline.sh` — and individual steps are documented in the README. API keys are read from `.env` (git-ignored) with a committed `.env.example`.

5 Conclusion

CourtEdge turns public NBA stats and sportsbook odds into a refined, monetizable data product for a specific, paying segment, defended by transparent methodology rather than hype. The technical core — distributed feature engineering over $\sim 599\text{k}$ play-by-play events, a calibrated win-probability model evaluated without leakage, and a unit-tested quant layer — is delivered through a deployed dashboard. We are honest about the system’s limits: a highly efficient market, a weak data moat, and a model that only modestly beats Elo. The value is not a magic edge but better prices, correct sizing, and an auditable CLV record.